Subject Description Form

Subject Code	ABCT5042
Subject Title	Quantitation and Reporting of Greenhouse Gases and Carbon Footprint and Climate Action Methodologies
Credit Value	6
Level	5
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	• Provides students with competence to assess, validate and verify statements on GHG to assure their quality, reliability, relevance, accuracy, reasonableness, and validity. It also aims to enable students to apply energy management system and environmental management system in methodologies for climate action. • Equips students with the skillset in establishing processes to develop, revise and manage methodologies on climate actions that support sustainability by mitigation of GHG. • Builds students' capability in quantitation and reporting of GHG inventory and carbon footprint (CFP) of a product. It also aims to enable students to apply life cycle assessment to quantify CFP of a product.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Appreciate international schemes on promoting carbon emission reduction and carbon credit trading (b) Apply the specification and the guidance for verification and validation of GHG statements (c) Interpret the competence criteria for GHG validation and verification bodies (d) Adopt the processes to develop, revise and manage methodologies on climate actions that support sustainability by mitigation of GHG to achieve carbon neutrality (e) Adopt energy management system and environmental management system to support climate action methodology (f) appreciate the international agreements relating to climate actions to achieve carbon neutrality (g) apply the theoretical principle and concept of life cycle of a product and its assessment methodologies, (h) master carbon auditing and the CFP quantification methodologies,

	(i) implement the procedure for quantitation and reporting of GHG emissions, (j) adopt relevant accounting protocol/guidance in GHG quantification and reporting
Subject Synopsis/ Indicative Syllabus	Topics to be included in this course: (a) Fundamental concepts and principles of quality management system (b) Requirements for verifying GHG statements relating to GHG inventories, GHG projects and carbon footprints of products (c) Principles and requirements for bodies performing validation and verification of environmental information statements including GHG statements, carbon footprints of products, environmental information for sustainability reporting, environmental information related to “green bond”, “climate finance” and other financial instruments, etc (d) Competence requirements for GHG validation teams and verification teams (e) International scheme on promoting carbon emission reduction – Science Based Target initiative and Climate Disclosure Project (f) Principles for establishing approaches and processes to identify, assess, revise methodologies on climate actions to address climate change including adaptation and mitigation (g) Ways to achieve carbon neutral and carbon trading market overview (h) International ESG reporting requirements - Global Reporting Initiative and International Sustainability Standards Board (i) International agreements relating to climate change such as The United Nations Framework Convention for Climate Change (UNFCCC), the Kyoto Protocol, the Paris Agreement, United Nations Sustainable Development Goal on Climate Actions, etc (j) The basic science of climate change due to GHG (k) Life cycle assessment (LCA) including life cycle inventory analysis, impact assessment, interpretation, and reporting (l) Principles and techniques for quantitation of CFP (m) Development of CFP per functional unit or partial CFP per declared unit, and issuance of the CFP report and GHG statement (n) Principles and requirements for designing, developing, managing and reporting organization-level GHG inventories including requirements for inventory quality management, reporting, internal audit, etc (o) Principles and requirements for determining baselines, and monitoring, quantifying and reporting of project emissions (p) Quantification, monitoring, and reporting of GHG emission reductions or removal enhancement (q) Accounting protocols/guidance for quantification and reporting of GHG inventory (r) Environmental labelling and declaration

Teaching/Learning Methodology

Interactive lectures with discussion in class and illustration of real cases will be used. This approach emphasizes student-centered learning.

Lectures supplemented with reading will be used to introduce the key concepts of the topics. Homework or assignments would be given to enhance students' learning. Students should complete the required readings before each class. Students should be prepared for briefing or discussing the guided reading in the class. Supplementary readings would be recommended to provide additional background and depth on specific areas of focus. Students will be engaged in experiential learning by working on assignments consisting of technical analysis projects. Students are expected to have basic experience using Microsoft Excel and basic quantitative skills. However, full Excel proficiency is not required. Guest speakers may be invited to give lectures on specific issues relating to greenhouse gas and climate action. Students would gain better understanding of how the knowledge and skills be applied for addressing real-life carbon accounting scenarios.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)									
		a	b	c	d	e	f	g	h	i	j
Written examination	20%	√	√	√	√	√	√	√	√	√	√
Home assignments	20%		√			√	√	√	√	√	
Quizzes and class exercises	10%	√		√	√		√				
Tests	20%	√	√			√			√	√	√
Group project presentation	30%		√		√			√	√	√	√
Total	100%										

All assessment methods are used to assess the students' understanding of the subject materials, while group project is also used to assess teamwork, presentation and communication skills. Continuous assessment may include quiz, class exercises, home assignments, etc.

Student Study Effort Expected

Class contact:	
▪ Lectures	78 Hrs.
Other student study effort:	
▪ Self-study	42 Hrs.
▪ Assignments, quiz and group project	120 Hrs.
▪ Total student study effort	240 Hrs.

Reading List and References	• ISO 9000, *Quality management systems – Fundamentals and vocabulary* • ISO 9001, *Quality management systems – Requirements* • ISO 50001, *Energy management systems – Requirements with guidance for use* • ISO 14001, *Environmental management systems – Requirements with guidance for use* • ISO 14064-3, *Greenhouse gases – Part 3: Specification with guidance for the validation and verification of greenhouse gas assertions* • ISO 14065, *Greenhouse gases – Requirements for greenhouse gas validation and verification bodies for use in accreditation or other forms of recognition* • ISO 14066, *Greenhouse gases – Competence requirements for greenhouse gas validation teams and verification teams* • ISO 14080, *Greenhouse gas management and related activities – Framework and principles for methodologies on climate actions* • ISO 14026, *Environmental labels and declarations – Principles, requirements and guidelines for communication of footprint information* • ISO 14027, *Environmental labels and declarations – Development of product category rules* • ISO 14040, *Environmental management – Life cycle assessment – Principles and framework* • ISO 14044, *Environmental management – Life cycle assessment – Requirements and guidelines* • ISO/TS 14071, *Environmental management – Life cycle assessment – Critical review processes and reviewer competencies: Additional requirements and guidance to ISO 14044:2006* • ISO 14064-1, *Greenhouse gases – Part 1: Specification with guidance at the organization level for quantification and reporting of greenhouse gas emissions and removals* • ISO 14064-2, *Greenhouse gases – Part 2: Specification with guidance at the project level for quantification, monitoring and reporting of greenhouse gas emission reductions or removal enhancements* • ISO 14067, *Greenhouse gases – Carbon footprint of products – Requirements and guidelines for quantitation* • *Science Based Target initiative* • IFRS S2 *Climate-related Disclosures* • Global Reporting Initiative